

Case Study Report: Google Play Store App Analysis

1. Executive Summary

The Google Play Store is one of the largest mobile application marketplaces, hosting thousands of applications across multiple categories. This project aims to analyze app characteristics such as ratings, installs, reviews, app size, content ratings, categories, genres, pricing models, and update trends.

Through Exploratory Data Analysis (EDA), several patterns were identified regarding app popularity, user satisfaction, developer behavior, and market trends. The analysis provides actionable insights for app developers, businesses, and marketers seeking to understand the mobile app ecosystem.

2. Project Objective

The primary objectives of this analysis were:

- Understand app rating distributions.
 - Analyze relationships between installs, reviews, ratings, and app size.
 - Identify the most popular app categories and genres.
 - Compare free and paid applications.
 - Study content rating distributions.
 - Investigate update frequency trends.
 - Discover characteristics of highly rated and highly reviewed apps.
 - Generate business insights to support app development strategies.
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3. Dataset Overview

The dataset contains information about Google Play Store applications, including:

Feature	Description
App	Application Name
Category	App Category
Rating	User Rating
Reviews	Number of Reviews
Size	App Size (MB)
Installs	Total Downloads
Type	Free or Paid
Price	Application Price
Content Rating	Target Audience
Genres	App Genre
Last Updated	Last Update Date

After data cleaning:

- Missing values were handled.
 - Installs were converted to numeric values.
 - Size values were standardized to MB.
 - Date columns were converted into datetime format.
 - Invalid records were removed.
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4. Data Cleaning Process

The following preprocessing steps were performed:

Missing Value Handling

- Removed rows with critical missing values.
- Filled or ignored non-critical missing data where appropriate.

Installs Conversion

- Removed commas and "+" signs.
- Converted installs into numeric format.

Size Standardization

- Converted KB values into MB.
- Removed textual symbols.

Date Formatting

- Converted "Last Updated" into datetime format.

Duplicate Removal

- Removed duplicate application records.

Outlier Treatment

- Invalid ratings and inconsistent values were checked.
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5. Exploratory Data Analysis

5.1 Distribution of App Ratings

Most applications received ratings between 4.0 and 4.5.

Key Findings

- Majority of apps have positive ratings.
- Very few apps have ratings below 3.0.
- Ratings are concentrated around 4.2.

Business Insight

Users generally provide favorable ratings, indicating overall satisfaction with apps available on the Play Store.

5.2 Relationship Between Reviews and Ratings

Analysis showed that apps with higher review counts tend to maintain stable ratings.

Key Findings

- Popular apps attract more reviews.
- High review volume increases rating reliability.

Business Insight

Businesses should focus on encouraging user reviews to build credibility and trust.

5.3 Distribution of App Sizes

App sizes show a right-skewed distribution.

Key Findings

- Most apps are below 20 MB.
- Few apps exceed 80 MB.
- Larger apps are less common.

Business Insight

Users generally prefer lightweight applications that consume less storage and data.

5.4 Free vs Paid Apps

Results

Type Count

Free 9,592

Paid 765

Key Findings

- Free apps dominate the marketplace.
- Paid apps represent a very small portion.

Business Insight

Freemium and ad-supported models are significantly more popular than paid-only models.

5.5 Content Rating Distribution

Major Content Ratings

- Everyone
- Teen
- Mature 17+
- Everyone 10+

Key Findings

- "Everyone" category dominates.
- Teen category ranks second.

Business Insight

Developers generally target broader audiences to maximize reach and downloads.

5.6 Top 5 Most Installed Apps

Applications such as Google Drive, Facebook, Instagram, and Google Play Games recorded the highest installs.

Key Findings

- Social and productivity apps dominate.
- Global brands attract billions of installs.

Business Insight

Strong brand recognition significantly impacts app adoption.

5.7 Apps Rated 4.0 and Above

Key Findings

- Majority of apps maintain ratings above 4.0.
- Only a small portion falls below 4.0.

Business Insight

Maintaining app quality directly influences user satisfaction and retention.

5.8 Average Reviews: Free vs Paid Apps

Key Findings

- Free apps receive substantially more reviews.
- Paid apps attract lower engagement.

Business Insight

Lower entry barriers increase user participation and review generation.

5.9 Average App Size by Category

Key Findings

- Game-related categories have the largest app sizes.
- Utility and business applications remain relatively smaller.

Business Insight

Entertainment apps often require richer multimedia resources.

5.10 Top Categories by Average Rating

Top-performing categories included:

- Events
- Education
- Art & Design
- Parenting
- Beauty

Business Insight

Specialized categories tend to achieve higher user satisfaction.

6. Intermediate Analysis

6.1 Top Categories by Installs

Top categories:

1. Game
2. Communication
3. Social
4. Productivity
5. Tools

Business Insight

These categories demonstrate the highest market demand and user engagement.

6.2 Content Rating Distribution: Free vs Paid Apps

Findings

- Both free and paid apps primarily target the "Everyone" audience.
- Teen-rated apps rank second.

Business Insight

Developers prioritize broader audience reach regardless of pricing strategy.

6.3 Most Reviewed Applications

Top reviewed apps include:

- Facebook
- WhatsApp Messenger
- Instagram
- Messenger
- Clash of Clans

Business Insight

Social media applications generate the highest user engagement globally.

6.4 App Update Trends

Findings

App updates increased dramatically after 2016.

2018 recorded the highest number of updates.

Business Insight

Frequent updates have become essential for app competitiveness, security, and user retention.

7. Advanced Analysis

7.1 Top 10 Highest Rated Apps

Findings

Several niche applications achieved perfect or near-perfect ratings.

However:

- Review counts remained relatively low.
- Install counts were significantly smaller than mainstream apps.

Business Insight

High ratings alone do not guarantee popularity.

7.2 Trend of App Updates Over Time

Findings

Update frequency increased steadily from 2010 to 2018.

Business Insight

Modern applications require continuous maintenance and feature improvements.

7.3 Average Rating by Install Range

Findings

Install Range	Avg Rating
0-1K	4.20
1K-10K	4.03
10K-100K	4.09
100K-1M	4.21
1M-10M	4.29
10M-100M	4.38
100M-1B	4.31

Business Insight

Higher-install apps generally maintain higher ratings.

7.4 Genre Analysis

Highest-rated genres include:

- Board;Pretend Play
- Comics;Creativity
- Puzzle;Education
- Strategy;Education

Business Insight

Creative and educational genres consistently deliver strong user satisfaction.

7.5 Genres with More Than 1 Million Installs

Top genres:

- Tools
- Action
- Photography
- Communication
- Productivity

Business Insight

Utility-focused and entertainment-focused genres dominate large-scale adoption.

8. Key Business Recommendations

For App Developers

- Prioritize user experience and performance.
- Maintain frequent updates.
- Encourage user reviews.
- Target high-demand categories.

For Product Managers

- Monitor ratings and reviews continuously.
- Focus on user retention strategies.
- Invest in app optimization.

For Marketing Teams

- Leverage social proof through reviews.
 - Promote highly rated features.
 - Use category-specific marketing campaigns.
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9. Conclusion

The Google Play Store ecosystem is dominated by free applications, high-demand categories such as Games, Communication, and Social, and apps that maintain strong user engagement through regular updates. The analysis demonstrates that popularity is influenced by a combination of installs, reviews, ratings, content accessibility, and continuous development efforts.

Apps with large install bases generally maintain higher ratings, suggesting a positive relationship between product quality and adoption. Additionally, categories focused on entertainment, productivity, and communication continue to dominate the market, while niche educational and creative genres achieve the highest user satisfaction scores.

Overall, the findings indicate that successful mobile applications combine strong functionality, frequent updates, broad audience appeal, and consistent user engagement strategies to sustain growth and competitiveness in the marketplace.